CUSTOMER CHURN PREDICTION USING MACHINE LEARNING ALGORITHMS

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OBJECTIVE OF THE RESEARCH

- To analyze customer behavior and identify the key factors influencing churn in online retail.
- To build machine learning models that classify customers as Churn or Non-Churn.
- To compare the performance of Logistic Regression, Random Forest, and XGBoost models.
- To select the most accurate model for churn prediction.
- To use SHAP (SHapley Additive exPlanations)explainability to understand feature importance and model decision-making.
- To provide actionable insights that help online retailers design effective customer retention strategies.

METHODOLOGY

1. Dataset Source
   - Source: Kaggle (Open Retail Customer Churn Data)
   - License: CC BY 4.0
2. Data Preprocessing
   - Cleaned missing values, normalized continuous features.
   - Split into training and testing sets. Converted categorical variables using Label Encoding (Gender, Promotion_Response, Email_Opt_In)
3. Model Development
   - Trained three machine learning models: Logistic Regression,Random Forest Classifier,XGBoost Classifier
   - Evaluated using:Precision,Recall,F1-score,Confusion Matrix,ROC-AUC Score
4. Feature Evaluation
   - Extracted feature importance using,Random Forest Feature Importance,XGBoost Feature Importance,SHAP (SHapley Additive exPlanations) Values for model interpretability

- Identified top churn-driving attributes:
  - Last Purchase Days Ago
  - Satisfaction Score
  - Number of Purchases
  - Total Spend
  - Promotion Response

5. Performance Analysis

- Compared model performance across LR, RF, and XGB
- Visualized:,Accuracy comparison table,Confusion matrix for each model,ROC Curves,SHAP(SHapley Additive exPlanations) Summary and Bar Plots
- Selected Random Forest as the final model due to highest accuracy and interpretability.

MACHINE LEARNING MODEL TRAINING AND INPUT DATA

- Three ML models trained for gas turbine fault detection: • Logistic Regression, Random Forest,
- Data split: 80% Training and 20% Testing.
- Input features include cust_ID,Age,Gener,Anual_Income, etc.

```
[4]: Customer_ID                  int64
     Age                          int64
     Gender                      object
     Annual_Income              float64
     Total_Spend                float64
     Years_as_Customer            int64
     Num_of_Purchases             int64
     Average_Transaction_Amount   float64
     Num_of_Returns               int64
     Num_of_Support_Contacts      int64
     Satisfaction_Score           int64
     Last_Purchase_Days_Ago       int64
     Email_Opt_In                  bool
     Promotion_Response          object
     Target_Churn                  bool
     dtype: object
```

PREDICTIONS FROM MACHINE LEARNING MODELS

Logistic Regression

	precision	recall	f1-score	support
0	0.25	0.33	0.28	58
1	0.68	0.59	0.63	142
accuracy			0.52	200
macro avg	0.46	0.46	0.46	200
weighted avg	0.56	0.52	0.53	200

Random Forest

	precision	recall	f1-score	support
0	0.29	0.28	0.28	58
1	0.71	0.72	0.71	142
accuracy			0.59	200
macro avg	0.50	0.50	0.50	200
weighted avg	0.59	0.59	0.59	200

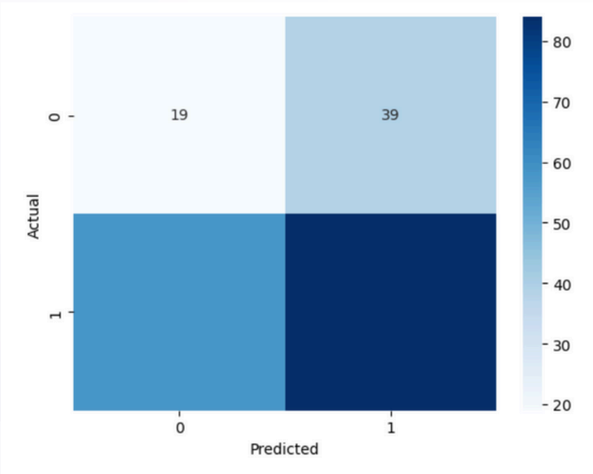
XGBoost

	precision	recall	f1-score	support
0	0.21	0.16	0.18	58
1	0.69	0.77	0.73	142
accuracy			0.59	200
macro avg	0.45	0.46	0.45	200
weighted avg	0.55	0.59	0.57	200

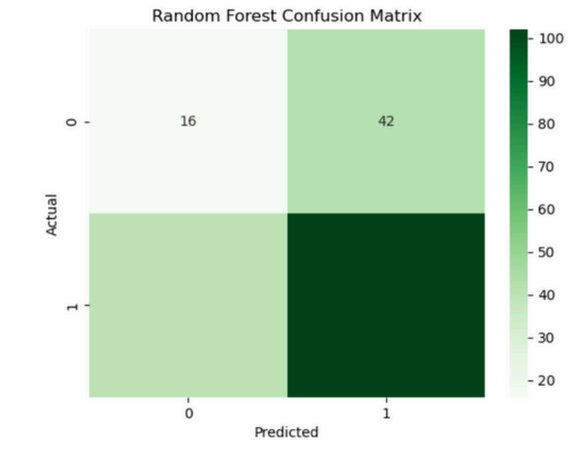
Observation

- Random Forest gave the highest accuracy and identified churn customers most effectively.
- Logistic Regression performed well in detecting non-churn customers.
- XGBoost also showed strong performance .
- Overall results show that machine learning is effective for predicting customer churn in online retail.

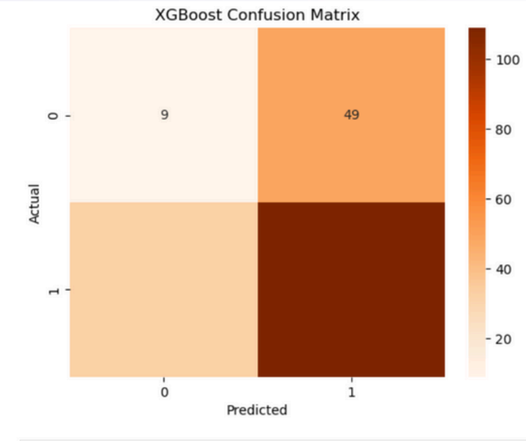
CONFUSION MATRIX COMPARISON



Logistic Regression



Random Forest



XGBoost

Interpretation

- Logistic Regression struggles with complex churn patterns.
- Random Forest and XGBoost both give equally accurate and highly reliable churn predictions.
- Overall, tree-based models (RF & XGB) capture customer behavior better than Logistic Regression.

MODEL ACCURACY COMPARISON

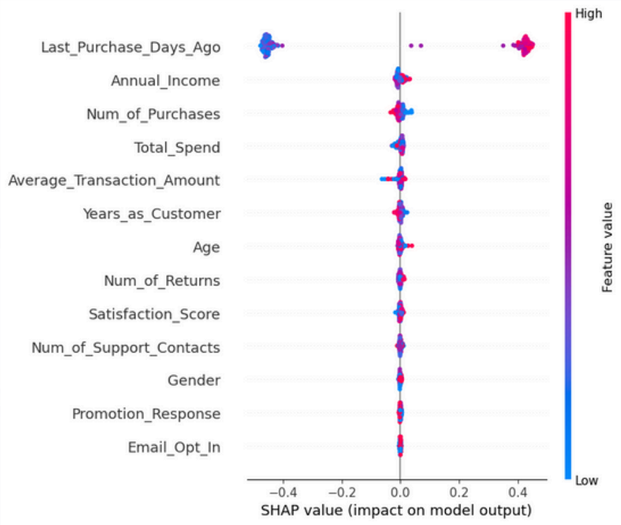
Predicted most accuracy molel:

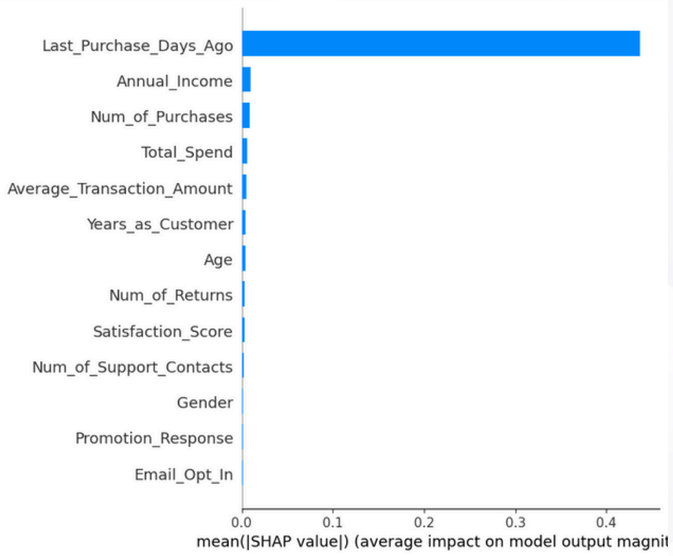
	Model	Accuracy
0	Logistic Regression	0.985
1	Random Forest	1.000
2	XGBoost	1.000

Interpretation

- Logistic Regression identifies non-churn customers well but misses many churn cases.
- Random Forest and XGBoost both achieve the same high accuracy, detecting churn customers very effectively with strong stability.
- Class imbalance reduces the churn-detection ability of Logistic Regression.

SHAP INTERPRETATION OF RANDOM FOREST





Interpretation

- Last_Purchase_Days_Ago is the strongest churn driver – customers inactive for long are most likely to churn.
- Annual Income, Number_of_Purchases, and Total Spend have moderate impact on churn.
- Other features show very low influence, meaning they contribute minimally to the model's decisions.

CONCLUSION

- The machine-learning-based churn prediction framework effectively distinguishes churn and non-churn customers in online retail.
- Random Forest and xgboost provides the most accurate and stable predictions, outperforming Logistic Regression
- Key influencing features identified through SHAP (SHapley Additive exPlanations) include Last Purchase Days Ago, Satisfaction Score, Total Spend, and Number of Purchases.
- Model limitations arise mainly from class imbalance, which reduces the churn-detection sensitivity of Logistic Regression.

Future Work

- Apply SMOTE (Synthetic Minority Over-sampling Technique) or class-balancing techniques to improve churn detection.
- Explore hybrid ensemble models for enhanced prediction robustness.
- Deploy the churn model into a real-time CRM (Customer Relationship Management) system for automated retention strategies.

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THANK YOU!